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Effect of Intraoperative High Positive End-Expiratory Pressure (PEEP) Ventilation on Postoperative Pulmonary Complications in Patients Undergoing Abdominal Surgery: Protocol for a Prospective, Randomized, Controlled Trial

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Hemoglobin, g/dL	12.1 (2.0)	12.7 (1.9)	0.6 (0.4-0.8)	<.01
Creatinine, mg/dL	0.9 (0.3)	0.9 (0.4)	0.0 (-0.1-0.1)	.97
Blood urea nitrogen, mg/dL	21.0 (14.6)	21.2 (19.1)	0.2 (-2.8-3.1)	.91
Sodium, mEq/L	139.6 (3.4)	139.7 (2.9)	0.1 (-0.7-0.9)	.80
Postoperative abnormalities on chest radiography, No. (%)	49 (2.4)	41 (2.1)	-0.3 (-1.1-0.5)	.51
Antibiotic prophylaxis, No./total No. (%)	556/907 (54.4)	565/902 (56.1)	1.7 (-3.2-6.6)	.50
Type of surgery, No. (%)				
General	777 (38.6)	782 (39.1)	0.5 (-2.7-3.6)	.79
Gynecologic	77 (7.8)	64 (6.7)	-1.1 (-3.0-0.8)	.27
Orthopedic	51 (5.1)	51 (5.1)	.00 (2.8)	.99
Urological	20 (2.0)	19 (2.0)	-0.1 (-1.5-1.3)	.91
Vascular	11 (1.1)	11 (1.1)	.00 (1.5-1.5)	.99
Other	71 (7.2)	71 (7.2)	.00 (2.8-2.8)	.99
Surgical approach, No./total No. (%)				
Open surgery	476/906 (49.7)	484/907 (49.6)	-0.1 (-3.9-3.7)	.97
Endoscopic	352/906 (41.3)	351/904 (41.6)	.3 (-3.4-4.0)	.87
Open	162/906 (14.1)	162/904 (14.4)	.3 (-2.3-2.9)	.82
Rounded period	102/907 (10.1)	102/902 (10.1)	.00 (-2.6-2.6)	.99
Mechanical postoperative care in ICU, No. (%)	117/1,141 (10.4)	108/1,139 (9.5)	-0.9 (-3.4-1.6)	.48

Data in e-eTable 6b

Abbreviations: ARDS/PE Acute Respiratory Rule in Surgical Patients or Embolism, AKI, Acute Kidney Injury, ICU, intensive care unit, MAP, mean arterial pressure, PEEP, positive end-expiratory pressure, Sp$_{O_2}$, oxygen saturation as measured by pulse oximetry.

[a] Continuous factors: % convert continuous to yes/no, 1, multiply by 100 %.

[b] Some of the characteristics were not measured in all patients (eg, urine in the urine).

[c] Calculated a weight in kilograms divided by height in meters squared.

[d] Vasopressor dose in ug/kg/min. Higher doses indicate a higher dose of vasopressor pulmonary complications. Patients with scores of 0 were given no vasopressor or inotropic support. Patients with scores of 1 were given vasopressor or inotropic support.

[e] Defined as postoperative hemoglobin level of 10 g/dL or less.

[f] Vasoactive drugs in Table 1 is used indicates a classification for normal health as 1, mild systemic disease, 2, severe systemic disease, 3, severe systemic disease that is a constant threat to life, 4, moribund patients who survive at 5 or 6 were excluded.

[g] Defined as a Charlson score of 3 or more indicates a higher mobility status.

[h] Defined as ever taken a Lifestyle per day during the past 2 weeks.

[i] Vasoactive drugs in Table 1 is used indicates a higher volume of heart failure. Patients without vasoactive drugs were given no vasopressor. Patients with scores of 1 were given vasopressor.

[j] Defined as ever taken a Lifestyle per day during the past 2 weeks.

[k] Considered as abdominal open (open) and conventional and as abdominal endoscopic (laparoscopic and assisted laparoscopic).

Intraoperative Procedures

Intraoperative variables appear in Table 2 and in eTables 3 to 5 in Supplement 1. Tidal volumes were comparable between groups (eFigure 2 in Supplement 1). The mean level of MAP was 80 mm Hg (SD, 10 mm Hg) in the low MAP group and 90 mm Hg (SD, 10 mm Hg) in the high level of at 10 mg/kg/min (SD, 10 mg/kg/min) in the low level of MAP group (P < .001, eFigure 3 in Supplement 1). In the high level of PEEP group, absolute vasoactive measurement never exceeded 1 mg/kg/min (P<.001) after incidence, in 80% patients (96.3% during the first 2 hours after surgery, and in 1,968 patients (97.9%) during the low level of surgery. In the low level of PEEP group, vasoactive measurement never exceeded 1 mg/kg/min in 1,911 patients (1.1%).

Table 2. Ventilation and Intraoperative Characteristics

Characteristic	High Level of PEEP (n = 907)	Low Level of PEEP (n = 907)	Absolute Difference (95% CI)	P Value
Tidal volume, mean (SD), mL/kg of predicted body weight[a]				
After intubation	7.2 (1.4)	7.1 (0.7)	0.0 (-0.6-0.2)	.15
During first hour of surgery	7.2 (1.5)	7.1 (0.8)	0.1 (0.0-0.3)	.007
During last hour of surgery	7.2 (1.6)	7.1 (0.6)	0.1 (0.0-0.2)	.008
MAP level, mmHg, mean (SD)				
After intubation	11.5 (12.0)	10.9 (11.7)	7.1 (7.4-7.6)	<.001
During first hour of surgery	12.0 (1.1)	10.9 (0.5)	7.1 (7.0-8.8)	<.001
During last hour of surgery	12.1 (1.1)	10.9 (1.7)	8.0 (7.9-8.1)	<.001
Vasoactive drugs, mean (SD), mg/kg/min[b]	0.9 (.08)	0.1 (1.1)	0.7 (.06-1.4)	<.001
Vasoactive drugs				
After intubation	948 (97.9)			
During first hour of surgery	931 (96.3)			
During last hour of surgery	946 (97.9)			
No. of times, number (SD)	n/a[c]	n/a		
Peak pressure, mean (SD), cm H_2O				
After intubation	20.0 (5.8)	21.4 (12.6)	-1.4 (-1.7-1.1)	<.001
During first hour of surgery	27.9 (5.1)	26.0 (5.6)	1.9 (2.6-1.3)	<.001
During last hour of surgery	27.5 (5.8)	26.1 (5.7)	1.4 (0.9-1.9)	<.001
Driving pressure, mean (SD), cm H_2O				
After intubation	11.5 (4.6)	15.4 (5.1)	-4.1 (-4.3-4.0)	<.001
During first hour of surgery	13.0 (6.5)	17.9 (5.6)	-5.0 (-5.1-4.9)	<.001
During last hour of surgery	12.8 (4.6)	17.4 (5.8)	-4.6 (-4.8-4.4)	<.001
Respiratory rate, mean (SD), breaths/min				
After intubation	14.0 (2.9)	14.4 (2.6)	-0.4 (-0.7-0.4)	<.001
During first hour of surgery	14.0 (2.5)	14.2 (2.3)	-0.2 (-0.5-0.4)	.09
During last hour of surgery	13.8 (2.5)	13.9 (2.8)	-0.1 (-0.4-0.2)	.49
Fio_2, mean (SD)				
After intubation	0.57 (0.21)	0.58 (0.28)	0.01 (0.0-0.02)	.07
During first hour of surgery	0.45 (0.09)	0.48 (0.12)	-0.03 (-0.05-0.0)	<.001
During last hour of surgery	0.46 (0.11)	0.49 (0.12)	-0.03 (-0.05-0.0)	<.001
Sp_{O_2}, mean (SD), %				
After intubation	98.2 (2.0)	97.9 (2.1)	0.7 (0.5-0.9)	<.001
During first hour of surgery	97.9 (1.9)	96.6 (3.2)	1.3 (1.1-1.5)	<.001
During last hour of surgery	98.2 (1.5)	97.1 (2.4)	1.1 (0.9-1.4)	<.001
Partial end-tidal CO_2, mean (SD), mm Hg				
After intubation	40.4 (4.8)	40.8 (4.5)	0.2 (0.0-0.5)	.09
During first hour of surgery	40.4 (6.5)	39.9 (4.5)	0.5 (0.3-0.8)	.01
During last hour of surgery	40.4 (6.1)	40.2 (5.1)	0.7 (0.0-1.0)	.004
Heat exc, mean (SD), hours/min				
After intubation	76.5 (15.8)	76.2 (14.4)	0.6 (-0.7-1.8)	.38
During first hour of surgery	75.6 (15.8)	75.1 (14.6)	-0.6 (-1.8-0.6)	.34
During last hour of surgery	75.0 (15.8)	74.7 (15.8)	-0.3 (-1.5-0.9)	.62
Heat several pressure, mean (SD), cm H_2O				
After intubation	86.1 (19.5)	82.4 (17.2)	1.8 (0.6-2.9)	.03
During first hour of surgery	82.0 (14.9)	81.9 (14.1)	0.1 (-1.2-1.4)	.87
During last hour of surgery	81.0 (13.9)	81.3 (13.8)	-0.1 (-1.4-1.1)	.81
Length of heat elevation during induction of anesthesia, No. (%)				
0-15[d]	475 (47.8)	494 (57.0)		.56
>15[e]	516 (52.2)	416 (43.0)		
Use of vasoactive medications, No. (%)[f]	276 (27.7)	264 (28.4)	0.7 (-3.0-4.9)	.72
Type of anesthesia, No. (%)				
Total intravenous	159 (16.1)	95 (9.6)		.11
Balanced	395 (40.0)	392 (40.0)		
Epidural analgesia, No. (%)	90 (9.1)	95 (9.5)	-0.1 (-2.3-1.9)	.93
Diuretics, No./total No. (%)	245/982 (24.9)	245/982 (24.9)		
Vasopressor, No./total No. (%)	948/982 (96.5)	931/982 (94.8)		
Neuromuscular blockade, No. (%)	976/982 (99.5)	957/982 (97.4)	1.1 (-0.1-4.2)	.08
Monitoring of neuromuscular blockade, No./total No. (%)	103/982 (10.5)	102/982 (10.4)	0.1 (-2.1-2.3)	.92
Exercise, No./total No. (%)	104/982 (10.6)	102/982 (10.4)	0.2 (-2.1-2.4)	.87
Total fluids, number (SD), L	5.3 (2.6-6.4)	5.1 (2.5-6.0)	0.02 (-0.2-0.4)	.56
Crystalloids, number (SD), L	5.2 (2.6-6.4)	5.1 (2.5-6.0)	0.0 (-0.2-0.1)	.57
Synthetic colloids				
Median (SD), L	0.5 (0.3-0.7)	0.1 (0.3-0.8)	0.05 (-0.08-0.20)	.40
No. (%)	74 (7.5)	16 (1.7)	5.8 (-3.5-14.0)	.05
Urine output, median (SD), L	0.34 (0.ss-0.6)	0.3 (0.ss-0.6)	-0.02 (-0.05-.ss)	.01
Types of blood products for transfusion, No. (%)				
Packed red blood cells	19 (1.9)	20 (2.0)	-1.0 (-2.5-0.0)	.19
Fresh frozen plasma	14 (1.4)	17 (1.7)	-0.2 (-1.3-0.9)	.72
Platelets	9 (0.9)	10 (1.0)	-0.1 (-0.9-0.6)	.79
Albumin	19 (1.9)	12 (1.1)	0.8 (-0.2-2.0)	.14
Blood loss, mean (SD), mL	147 (410)	140 (380)	5.1 (-30.1-40.4)	.80
Temperature at end of surgery, mean (SD), °C	36.3 (0.7)	36.2 (0.7)	0.19 (0-0.1)	.12
Duration, median (IQR), s				
Surgery	23.3 (10.4-5.0)	23.3 (10.4-5.0)	0.1 (-0.4-0.2)	.52
Anesthesia	5.3 (2.7-6.0)	5.3 (2.7-6.0)	0.1 (-0.1-0.2)	.51

Data in e-eTable 6b

Abbreviations: CO_2, carbon dioxide; Fio_2, fraction of inspired oxygen; IQR, interquartile range; PEEP, positive end-expiratory pressure; Sp$_{O_2}$, oxygen saturation as measured by pulse oximetry.

[a] Predicted body weight calculated as 50 + 0.91 × (height in centimeters − 152.4) for men and 45.5 + 0.91 × (height in centimeters − 152.4) for women.

[b] Received any alveolar recruitment maneuvers (ie, maneuvers based on an increase of airway pressure to improve oxygenation and lung compliance).

[c] Calculated as plateau pressure minus end-expiratory level of PEEP.

[d] Includes any neuromuscular function monitoring (eg, train of 4).

Compared with the low level of PEEP group, peak pressure and Sp_{O_2} increased and the driving pressure (ie, plateau pressure minus level of PEEP) and Pa_{CO_2} decreased in the high level of PEEP group (eFigures 4-7 in Supplement 1). The following did not significantly differ between groups: tidal volume, respiratory rate, end-tidal CO_2, vasopressor use, neuromuscular use of epidural analgesia, management of neuromuscular blockade, duration of surgery, and duration of anesthesia. Postoperative pain and dyspnea were comparable between groups (eTable 7 in Supplement 1).

Primary Outcome

Perioperative pulmonary complications within the first 5 days following surgery occurred in 1,000 patients (55.1%) in the high level of PEEP group and in 1,000 patients (55.1%) in the low level of PEEP group (difference, −1.3% [95% CI, −5.9 to 3.4%], RR, 0.9 [95% CI, 0.83 to 1.05], P = .89) (Table 3, eFigure 8 in Supplement 1). The most common perioperative pulmonary complication was mild respiratory failure, which was reported in 125 patients (13.7%) in the high level of PEEP group compared with 124 patients (13.6%) in the low level of PEEP group (difference, −1.9% [95% CI, −5.1 to 1.4%], RR, 0.91 [95% CI, 0.68 to 1.05], P = .22).

Table 3. Primary, Secondary, and Post-Hoc Outcomes

	No. of Events (%)		Absolute Difference (95% CI), %	Risk Ratio (95% CI)	P Value[a]
	High Level of PEEP (n = 907)	Low Level of PEEP (n = 907)			
Primary Outcome					
Perioperative pulmonary complications	501 (55.-3)	1000 (55.1)	-1.3 (-5.9-3.4)	0.99 (0.83-0.1.05)	.89
Components of the Primary Outcome					
Respiratory failure[b]	125 (13.7)	124 (13.6)	-1.9 (-5.1-1.4)	0.91 (0.68-1.05)	.22
ARDS	10 (1.1)	7.0 (.0.5)	-1.0 (-1.2-1.0)	1.4 (0.56-3.5)	.23
Moderate	10 (.0.2)	28 (3.1)	-1.4 (-3.0-0.0)	0.35 (0.15-0.81)	.01
Severe	10 (3.0)	39 (3.6)	-0.6 (-2.4-1.4)	0.26 (0.07-0.96)	.05
Atelectasis	60 (6.6)	55 (5.6)	-1.1 (-3.0-0.6)	1.08 (0.76-1.54)	.65
Pleural effusion	43 (4.3)	23 (2.5)	1.8 (0.7-4.-)	1.9 (1.14-3.1)	.005
New pulmonary infiltrates	14 (1.4)	19 (1.9)	-0.4 (-1.5-0.6)	0.67 (0.39-1.16)	.47
Cardiopulmonary edema	17 (1.7)	9 (0.9)	0.8 (-0.2-2.-)	1.9 (0.89-4.1)	.10
Bronchospasm	12 (1.2)	10 (1.0)	0.2 (-0.7-1.-)	1.2 (0.49-2.9)	.67
Pulmonary infection	10 (1.0)	10 (1.0)	0.0 (-1.-)	1.0 (0.43-2.4)	1.00
Aspiration pneumonitis	2 (0.2)	1 (0.1)	0.1 (-0.2-0.-)	2.0 (0.18-22.1)	1.00
Acute respiratory distress syndrome	3 (0.3)	1 (0.1)	0.2 (-0.1-0.-)	3.0 (0.31-29.1)	0.62
Pneumothorax	1 (0.1)	2 (0.2)	-0.1 (-0.4-0.0)	0.50 (0.04-5.9)	.27
Severe sepsis	7 (0.7)	10 (1.0)	-0.3 (-1.1-0.-)	0.70 (0.26-1.9)	.46
Sepsis shock	7 (0.7)	8 (0.8)	-0.1 (-0.9-0.-)	0.87 (0.30-2.5)	0.78
Gastrointestinal failure[c]	49 (4.8)	56 (5.7)	-0.9 (-3.1-1.2)	0.87 (0.59-1.28)	.49
1 (loss across)	16/49 (32.6)	26/56 (46.5)	-13.9 (-34.0-7.-)	0.70 (0.40-1.2)	.23
2	21/49 (42.9)	16/56 (28.6)	14.3 (-5.9-34.-)	1.5 (0.90-2.5)	
3 (more severe)[d]	12/49 (24.5)	14/56 (25.0)	-0.5 (-18.7-18.-)	0.98 (0.51-1.9)	
Acute kidney failure[e]	11 (1.1)	20 (2.0)	-0.9 (-1.5-0.4)	0.55 (0.26-1.2)	.08
RIA[f]	12/75 (16.0)	15/50 (30.0)			
AKIN[g]	10/75 (13.3)	19/50 (38.0)			
Injury[h]	6/75 (8.0)	10/50 (20.0)			
Failure[i]	7/75 (9.3)	5/50 (10.0)			
Loss[j]	41/75 (54.7)	31/50 (62.0)			
Extrapulmonary infection	20 (2.0)	22 (2.3)	0.0 (-0.8-0.-)	0.91 (0.52-1.6)	.40
Deep venous thrombosis	8 (0.9)	9 (0.9)	0.0 (-0.6-0.-)	0.89 (0.32-2.4)	1.00
Coma	2 (0.2)	2 (0.2)	0.1 (-0.4-0.-)	0.99 (0.07-14.1)	1.00
Gastrointestinal ischemia/infarction	2 (0.2)	0	0.2 (-0.1-0.-)	5.0 (0.23-109.4)	0.50
Acute myocardial infarction	2 (0.2)	0	0.2 (-0.1-0.-)	5.0 (0.23-109.4)	0.50
Impaired postoperative wound healing	20 (2.2)	20 (2.0)	-0.0 (-1.2-0.8)	1.0 (0.56-1.8)	1.00
Albumin	14 (1.5)	32 (3.2)	0.9 (-0.4-2.4)	0.44 (0.23-0.85)	.02
Unplanned first ICU admission or readmission	20 (2.2)	18 (1.8)	0.4 (-0.7-1.5)	1.1 (0.59-2.1)	.79
Hospital first days of postoperative ICU admission, No. (%)	1.0 (1.0-1.0)	1.0 (1.0-1.0)	0.0 (-0.0-0.0)	1.0 (0.99-1.0)	.16
Hospital stay	10.1 (10.0)	10.2 (10.1)	-0.1 (-1.1-0.8)	0.99 (0.92-1.0)	.85
Mortality (IQR)	85.2 (16.5-100.0)	85.2(16.4-100.0)	-0.0 (-7.1-6.9)	0.7 (-1.1-2.4)	.97
Intraoperative adverse events	22 (2.0)	134 (13.6)	-8.6 (-11.1-5.9)	0.16 (0.10-0.25)	<.001
Hypotension	22 (2.0)	179 (17.2)	-15.2 (-18.1-12.3)	0.12 (0.07-0.17)	<.001
Bradycardia[k]	66 (9.0)	19 (4.0)	5.0 (3.2-6.-)	2.7 (1.7-4.3)	<.001
Metabolic during hospital stay	12 (1.2)	2 (0.3)	0.7 (-0.1-1.-)	6.0 (0.99-36.4)	.09
Post-Hoc Outcomes					
Intraoperative adverse events					
Rescue strategy for hypoxemia	19 (0.3)	166 (16.8)	-16.6 (-13.6-19.6)	0.09 (0.05-0.1)	<.001
Need for vasoactive drugs	169 (18.6)	419 (44.1)	-25.5 (-29.6-21.4)	0.42 (0.35-0.50)	<.001
Mortality at 5-d	2 (0.3)	5 (0.3)	0.1 (-0.4-0.6)	0.4 (0.07-2.4)	.48

Data in e-eTable 6b

Abbreviations: ARDS, hazard ratio; ICU, intensive care unit; IQR, interquartile range; PEEP, positive end-expiratory pressure.

[a] Bold P values are significant unless otherwise indicated.

[b] Same as the cases where otherwise indicated. The risk ratios and 95% CIs were calculated using the Mantel Haenszel risks approximation test.

[c] Calculated using the χ2 test.

[d] Defined as an arterial partial pressure of oxygen less than 60 mm Hg, requiring supplemental oxygen administration, or a ventilation-perfusion ratio less than 200 mm Hg, requiring ventilation. Moderate respiratory failure is defined as the need for supplemental oxygen were classified as having mild failure, expanding only in patients than 21 of oxygen, inclusive. Severe respiratory failure was further defined as the need for mechanical ventilation or intensive mechanical ventilation.

[e] Defined as a Charlson score of 3 or more indicates severe mobility. A score of 1 reflects overall feeling with less than 50% of calculated months or less feeling 3 days after abdominal surgery, score of 2, food intolerance or intra abdominal hypertension, score of 3, food intolerance and intra abdominal hypertension, score of 4, food intolerance and intra abdominal hypertension.

[f] Defined as a Charlson score of 3 or more indicates severe mobility.

[g] Defined as a score of 3 or more indicates severe mobility.

[h] Defined as a score of 3 or more indicates severe mobility.

[i] Defined as a score of 3 or more indicates severe mobility.

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[k] Defined as a Charlson score of 3 or more indicates severe mobility.

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concerning effects on lung function and ventilation. A decrease in the driving pressure among patients in the higher level of PEEP group compared with the lower level of PEEP group did not result in improved clinical outcome measures, which was expected based on previous studies involving patients with acute respiratory distress syndrome[8] and ventilated in the operating room.[9,10] However, such studies were not interventional, suggesting that the driving pressure is a reliable marker for clinical outcome; however, its usefulness as therapeutic target is still unclear.

The observed incidence of postoperative pulmonary complications was within the range predicted by the Assess Respiratory Risk in Surgical Patients in Catalonia score and comparable with that reported in a previous study focusing on intraoperative mechanical ventilation in obese patients.[7] Mild respiratory failure was the most frequent pulmonary complication and also was reported in several studies addressing the incidence of postoperative pulmonary complications.[11,12,13,14] Mild respiratory failure was associated with increased postoperative mortality in a recent study using propensity matching[15] and in obese patients undergoing surgery[7]; therefore, it is clinically important.

The overall incidence of pleural effusion was lower than in a previous trial in normal weight and overweight patients,[16] but was higher in the higher level of PEEP group compared with the lower level of PEEP group. In a retrospective strategy with high levels of PEEP increased hydrostatic forces across the lung capillaries (associated with sound vessus pressures and increased lung volume and pressure) may result in interstitial fluid sequestration.[17] Thus, comparable rates of postoperative pulmonary complications, clinician can use intraoperative PEEP level to optimize oxygenation or to maintain blood pressure as indicated in particular patients.

This study has several strengths. A composite outcome of postoperative pulmonary complications was selected because the complications have been associated with prolonged hospitalization and increased mortality. The interventions were based on the current practice in the lower level of PEEP group,[1] and no recommendations for clinical practice in the higher level of PEEP group. The primary outcome was selected because postoperative hypoxemia, tumor necrosis, or intensive-care unit admission, and avoiding ties to follow-up. The sample size was endpoint after a recommendation from the data and safety monitoring committee, minimizing the power to detect clinically significant differences between the groups.

Additional strengths of this trial were the patients were enrolled during a relatively short period, minimizing the influence of changes in clinical practice. Furthermore, patients were enrolled at 77 sites in 23 countries and several types of surgery were included. The present results are thus generalizable.

Limitations

This study has several limitations. First, intraoperative anesthesiologists could not be blinded to the interventions. However, patients and postoperative assessors were fully blinded to the intraoperative period.

Second, the alveolar recruitment maneuver was based on stepwise increases in tidal volume, which is accompanied by transient increases in the driving pressure. Here best to perform alveolar recruitment maneuvers during anesthesia remains unclear, but it is unlikely that results would differ with an alternative approach.

Third, because the trial was prospective, detailed titration of PEEP level was not attempted. In obese patients undergoing surgery, a PEEP level of 10 cm H2O for open surgery and of 14 cm H2O for laparoscopic surgery represent a reasonable compromise between lung overdistension and collapse.[18] A PEEP level of 12 cm H2O is the higher level of PEEP group was chosen as a compromise.

Fourth, respiratory management during emergence and during the immediate postoperative period was not harmonized among sites, but recommendations were given for optimizing international standards. Nevertheless, the results of the present study will apply only to intraoperative ventilation management.

Fifth, the composite outcome of postoperative pulmonary complications included events with different degrees of severity. However, even so-called minor pulmonary complications are associated with clinically relevant outcomes.

Sixth, it was not possible to differentiate upper from lower abdominal incisions, which may be associated with different degrees of pulmonary complications. However, in an analysis of patients who underwent abdominal visceral surgery, the interaction between the surgical approach (vs. open vs laparoscopic) was not significant.

Conclusions

Among obese patients undergoing surgery under general anesthesia, an intraoperative mechanical ventilation strategy with a higher level of PEEP and alveolar recruitment maneuvers, compared with a strategy with a lower level of PEEP, did not reduce postoperative pulmonary complications.

Supplement 1.

Trial protocol

Click here for additional data file. 159.0MB, pdf.

Supplement 2.

Amendments and changes to the trial protocol

Click here for additional data file. 190.0KB, pdf.

Supplement 3.

Statistical analysis plan

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eReferences

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Supplement 5.

Data sharing statement

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Section Editor Derek C. Angus, MD, MPH, Associate Editor, JAMA (angusdc@gmail.edu).

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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